

<i>index, n, i, j, k</i>	index metavariables
<i>int, v</i>	integer literal
<i>ident, x</i>	variable identifiers
<i>fident, f</i>	function identifier
<i>tvar, T</i>	type-level variable identifiers

lin	$::=$ $ \quad n$ $ \quad T$ $ \quad lin + lin'$ $ \quad lin - lin'$ $ \quad n * T$	Linear arithmetic expression integer literal integer variable linear addition linear subtraction linear product
pow, p	$::=$ $ \quad 1$ $ \quad 2$ $ \quad 2 \wedge lin$	Power of two one two linear exponent of two
$size, s$	$::=$ $ \quad n$ $ \quad p$ $ \quad s/p$ $ \quad s + s'$ $ \quad s - s'$ $ \quad s * s'$	Sizes as dyadic numbers constant size power of two size division of size power of two addition of sizes subtraction of sizes multiplication of sizes
$range, R$	$::=$ $ \quad s..s'$ $ \quad s, p..s'$	range of numeric indices Range with unit step Range with step power of two
$tkind, K$	$::=$ $ \quad \mathbf{Group}$ $ \quad \mathbf{Field}$ $ \quad \mathbf{Size}$	Type kinds Elliptic curve group Finite field modulo prime Size type variable
$tbase, B$	$::=$ $ \quad \mathbf{Scalar} \ T$ $ \quad \mathbf{Pairing} \ T_1 \ T_2$ $ \quad \mathbf{Additive} \ T$ $ \quad \mathbf{Multiplicative} \ T$ $ \quad T$	Base types Scalar field of group Pairing-friendly groups Additive subgroup of field Multiplicative subgroup of field Type variable
$type, t$	$::=$ $ \quad \mathbf{Uni} \langle B, s \rangle$ $ \quad \mathbf{Mle} \langle B, s \rangle$ $ \quad [t; s]$ $ \quad \mathbf{Fin} \langle R \rangle$ $ \quad B$	Static types Univariate polynomial Multilinear polynomial Vector type Index type Base type
$tdecl, TD$	$::=$ $ \quad T : K$	Type variable of given kind
$tdecls, TDs$	$::=$ $ \quad TD_0, \dots, TD_n$	Type variable declarations

$qualifier, q$	$::=$ $ $ public $ $ private	Visibility qualifier Publicly visible argument Private argument
arg	$::=$ $ $ $q\ x : t$	Argument Typed, qualified argument
$args$	$::=$ $ $ $arg_0, \dots, arg_n$	Arguments Argument list
bin_op, op	$::=$ $ $ $+$ $ $ $-$ $ $ $*$ $ $ $\cdot$ $ $ $/$ $ $ $\wedge$ $ $ $++$	Zippel binary operators Addition Subtraction Product Inner product Division Exponentiation Vector concatenation
exp, e, d	$::=$ $ $ v $ $ x $ $ $e\ op\ d$ $ $ gen $\langle T \rangle$ $ $ coef (e) $ $ mle (e) $ $ interpolate (e, d) $ $ random $\langle t \rangle$ $ $ challenge $\langle t \rangle$ $ $ $[e_0, \dots, e_n]$ $ $ $[e\ \text{for } x\ \text{in } r]$ $ $ $x[s]$ $ $ $f(e_0, \dots, e_n)$ $ $ reduce $(op\ \text{"}, "$ aexp $)$ $ $ assert $(b); e$ $ $ verify $(b); e$ $ $ let $x := e\ \text{in } d$ $ $ $x \leftarrow e; d$	Zippel expressions Field literal Variable Binary operation Group generator Univariate polynomial coefficients Multilinear evaluations over bool h Lagrange interpolation from points Sample pseudorandom number gen Generate a verifier challenge Vector of expressions Map comprehension index a vector Function application Reduce binary operation over vect Prover assertion Verifier assertion Let binding Transcript binding
$bexp, b$	$::=$ $ $ $e = e'$ $ $ $f(e_0, \dots, e_n)$ $ $ $b \ \&\& \ b'$ $ $ $b \ \ b'$	Boolean expressions Equality of expressions Protocol application Boolean and operation Boolean or operation
$decl, D$	$::=$ $ $ proto $f \langle TDs \rangle (args)\ \text{where } b := e; \text{verify } (b').$ $ $ $f_n f \langle TDs \rangle (args) \rightarrow t := e.$	Top-level declarations Zippel protocol Zippel function